WEEK 4

Sets and Set Operations

Objectives

By the end of this week, you should be able to:

- Identify the types of sets
- Find the Power set of the given set
- Find the Cartesian product of two sets
- Perform various set operations

4.1 Sets

A set is an unordered collection of objects which are called elements or members. We can denote a set S by listing its elements in curly braces.

The following are some examples of a set:

Example 4.1. *The set V of all vowels in the English alphabet is $V = \{a, e, i, o, u\}$.*

Example 4.2. *The set O of odd positive integers less than 10 is $O = \{1, 3, 5, 7, 9\}$.*

We can also denote a set S by

- listing some of its elements in curly braces. Members of the set are listed, and then ellipses $(\ldots)$ are used when the general pattern of the elements is obvious.

Example 4.3. *The set of positive integers less than 100 can be denoted by* $\{1, 2, 3, \ldots, 99\}$.

- using set builder notation. Elements in the set are characterized by stating their property or properties.

Example 4.4. *The set* O *of odd positive integers less than 10 can be written as* $O = \{x | x \text{ is an odd positive integer less than } 10\}$.

We can also denote a set S graphically using Venn Diagrams.

- Represent a universal set U (set containing all objects under consideration) by the rectangle.
- Use circles inside the rectangle to represent sets.

Venn Diagrams are used to indicate relationships between sets.

Example 4.5. *Draw a Venn diagram that represents* V*, the set of vowels in the English alphabet.*

Some Important Sets

- $\mathbf{N} = \{0, 1, 2, 3, \ldots\}$, the set of natural numbers.
- $\mathbf{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, the set of integers.
- $\mathbf{Z}^+ = \{1, 2, 3, \ldots\}$, the set of positive integers.
- $\mathbf{Q} = \{p/q \mid p \in \mathbf{Z}, q \in \mathbf{Z}, \text{ and } q \neq 0\}$, the set of rational numbers.
- $\mathbf{R}$, the set of real numbers.
- $\mathbf{R}^+$, the set of positive real numbers.
- $\mathbf{C}$, the set of complex numbers.

4.1.1 Empty Sets

This is a special set that has no elements. It is also called a null set and is denoted by $\varnothing$ or $\{\ \}$.

Example 4.6. *The set of all positive integers that are greater than their squares is a null set.*

4.1.2 Finite Set

Let S be a set. If there are exactly n distinct elements in S where n is a non-negative integer, we say that S is a finite set and that n is the cardinality of S. The cardinality of S is denoted by $|S|$.

Example 4.7. *Let A be the set of odd positive integers less than 10. Then A is a finite set of cardinality $|A| = 5$.*

4.1.3 Infinite Set

A set is said to be infinite if it is not finite.

Example 4.8. *The set of positive integers is infinite.*

4.1.4 Equal Set

Two sets are equal if and only if they have the same elements.

Example 4.9. *The sets $\{1, 3, 5\}$ and $\{3, 5, 1\}$ are equal, because they have the same elements. Note that the order in which the elements of a set are listed does not matter.*

If A and B are sets, then A and B are equal denoted as $A = B$ if and only if $\forall x\ (x \in A \leftrightarrow x \in B)$ is true.

4.1.5 Subset

The set A is said to be a subset of B if and only if every element of A is also an element of B. We use the notation $A \subseteq B$ to indicate that A is a subset of set B.

Example 4.10. *The set of all odd positive integers less than 10 is a subset of the set of all positive integers less than 10.*

We see that $A \subseteq B$ if and only if the quantification $\forall x\ (x \in A \to x \in B)$ is true.

Note: Every nonempty set S is guaranteed to have at least two subsets, the empty set and the set S itself, that is, $\varnothing \subseteq S$ and $S \subseteq S$.

When we wish to emphasize that a set A is a subset of a set B but that $A \neq B$, we write $A \subset B$ and say that A is a proper subset of B. For $A \subset B$ to be true, it must be the case that $A \subseteq B$ and there must exist an element x of B that is not an element of A.

Example 4.11. *Let* $A = \{0, 1, 2\}$ *and* $B = \{0, 1, 2, 3\}$*, then* $A \subseteq B$*.*

4.2 The Power Set

Given a set S, the power set of S is the set all subsets of S. The power set of S is denoted by $P(S)$.

Example 4.12. *The set* $A = \{3\}$ *has exactly two subsets, namely,* $\varnothing$ *and the set* $\{3\}$ *itself. Therefore,* $P(A) = \{\varnothing, \{3\}\}$*.*

Example 4.13. *The set* $B = \{1, 3\}$ *has exactly four subsets, namely,* $\varnothing$*,* $\{1\}$*,* $\{3\}$ *and* $\{1, 3\}$*. Therefore,* $P(B) = \{\varnothing, \{1\}, \{3\}, \{1, 3\}\}$*.*

Example 4.14. *Given the set* $S = \{0, 1, 2\}$*. The power set* $P(S)$ *is the set of all subsets of* $\{0, 1, 2\}$*. Hence,*

$$P(S) = \{\varnothing, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}$$

4.3 Cartesian Products

The cartesian products of sets A and B, denoted by $A \times B$ is the set of all ordered pairs (a, b), where $a \in A$ and $b \in B$. Hence

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}.$$

Example 4.15. *Given* $A = \{1, 2\}$ *and* $B = \{a, b, c\}$*. The Cartesian product* $A \times B$ *is*

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

The Cartesian product $B \times A$ *is*

$$B \times A = \{(a, 1), (b, 1), (c, 1), (a, 2), (b, 2), (c, 2)\}$$

4.4 Set Operations

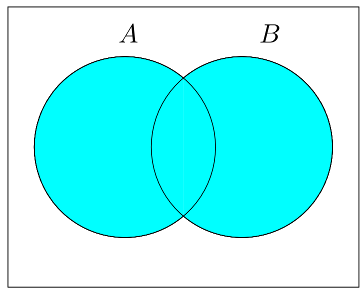
We can create new sets by combining two or more sets in different ways.

4.4.1 The Union of Sets

For sets A and B, their union $A \cup B$ is the set containing all elements that are either in A, or in B or in both.

$$A \cup B = \{x | x \in A \vee x \in B\}.$$

Venn diagram: The shaded region represents $A \cup B$.



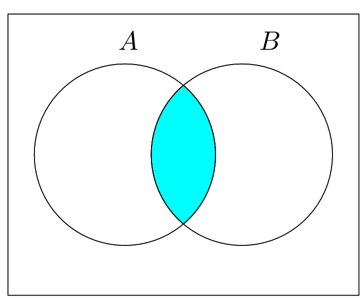
Example 4.16. *If* $A = \{0, 1, 2\}$ *and* $B = \{1, 2, 3, 5\}$, *then* $A \cup B = \{0, 1, 2, 3, 5\}$.

4.4.2 The Intersection of Sets

For sets A and B, their intersection $A \cap B$ is the set containing those elements that are both in A and B.

$$A \cap B = \{x | x \in A \wedge x \in B\}.$$

Venn diagram: The shaded region represents $A \cap B$.

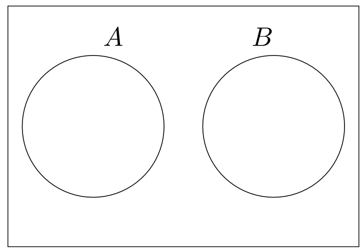


Example 4.17. *If* $A = \{0, 1, 2\}$ *and* $B = \{1, 2, 3, 5\}$, *then* $A \cap B = \{1, 2\}$.

4.4.3 Disjoint Set

Two sets A and B are called disjoint iff their intersection is an empty set. ($A \cap B = \varnothing$).

Venn diagram:



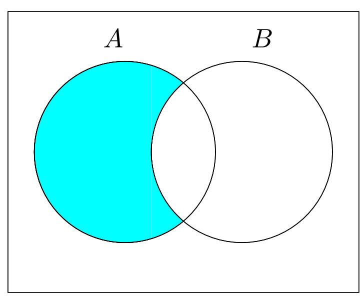
Example 4.18. *$A = \{1, 3, 5, 7, 9\}$ and $B = \{2, 4, 6, 8, 10\}$ are disjoint because $A \cap B = \varnothing$.*

4.4.4 Set Difference

For sets A and B, the difference of A and B, written as $A - B$, is the set of all elements that are in A but not in B.

$$A - B = \{x | x \in A \wedge x \notin B\}.$$

Venn diagram: The shaded region represents $A - B$.



Example 4.19. *If $A = \{0, 1, 2\}$ and $B = \{1, 2, 3, 5\}$, then*

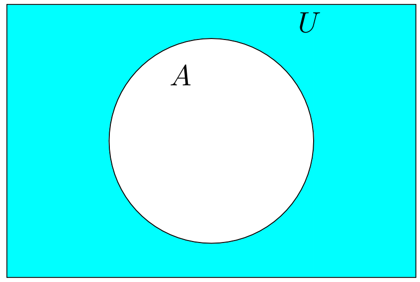
- *$A - B = \{0\}$ and*
- *$B - A = \{3, 5\}$.*

4.4.5 Set Complements

The complement of A, written as $\overline{A}$, is the difference of the universal set U and A, i.e., $\overline{A} = U - A$.

$$\overline{A} = \{x | x \notin A\}.$$

Venn diagram: The shaded region represents $\overline{A}$.



Example 4.20. *Find the complement of A if the universal set is the set of all positive integers.*

1. $A = \{1, 2\}$.
2. $A = \{10, 11, 12, \ldots\}$.

Solution:

1. $\bar{A} = \{3, 4, 5, \ldots\}$.
2. $\bar{A} = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

4.5 Membership Table

A membership table is similar to the truth table except we use 1 and 0 instead of T and F. 1 indicates that an element is in the set and 0 indicates that an element is not in the set.

4.5.1 Membership Table for $\overline{A}$

Similar to the truth table for negation operator.

A	$\overline{A}$
1	0
0	1

4.5.2 Membership Table for $A \cup B$

Similar to the truth table for “or" operator.

A	B	$A \cup B$
1	1	1
1	0	1
0	1	1
0	0	0

Example 4.21. *Construct the membership table for*

1. $A \cup \overline{B}$.
2. $\overline{A \cup B}$.

4.5.3 Membership Table for $A \cap B$

Similar to the truth table for “and" operator.

A	B	$A \cap B$
1	1	1
1	0	0
0	1	0
0	0	0

Example 4.22. *Construct the membership table for*

1. $A \cap \overline{B}$.
2. $\overline{A \cap B}$.

4.5.4 Membership Table for $A - B$

Note that $A - B = A \cap \overline{B}$. So the membership table of $A - B$ is same as the membership table for $A \cap \overline{B}$.

A	B	$A - B$
1	1	0
1	0	1
0	1	0
0	0	0

Example 4.23. *Construct the membership table for* $B - A$.

More Examples

Construct the membership table for the following:

1. $A \cup (B \cap C)$
2. $A \cap (\overline{B \cap C})$
3. $(A \cup B) - C$

4.5.5 Proving Set Identities Using a Membership Table.

Let S and T be sets. To show that $S = T$, we need to show that the columns for S and T have the same values.

Example 4.24. *Show that* $\overline{A \cup B} = \overline{A} \cap \overline{B}$.

Solution: We can show this by using a membership table.

A	B	$A \cup B$	$\overline{A \cup B}$	$\overline{A}$	$\overline{B}$	$\overline{A} \cap \overline{B}$
1	1	1	0	0	0	0
1	0	1	0	0	1	0
0	1	1	0	1	0	0
0	0	0	1	1	1	1

Since the columns for $\overline{A \cup B}$ and $\overline{A} \cap \overline{B}$ are the same, we conclude that $\overline{A \cup B} = \overline{A} \cap \overline{B}$.

Example 4.25. *Show that* $(A \cup B) - (C - A) = A \cup (B - C)$.

Solution: We can show this by using a membership table.

A	B	C	$A \cup B$	$C - A$	$(A \cup B) - (C - A)$	$B - C$	$A \cup (B - C)$
1	1	1	1	0	1	0	1
1	1	0	1	0	1	1	1
1	0	1	1	0	1	0	1
1	0	0	1	0	1	0	1
0	1	1	1	1	0	0	0
0	1	0	1	0	1	1	1
0	0	1	0	1	0	0	0
0	0	0	0	0	0	0	0

Since the columns for $(A \cup B) - (C - A)$ and $A \cup (B - C)$ are the same, we conclude that $(A \cup B) - (C - A) = A \cup (B - C)$.

4.5.6 Proving Set Identities Using Set Builder Notations.

We can use set builder notation and logical equivalences to establish set identities. Recall that:

- $A \cup B = \{x | x \in A \vee x \in B\}$.
- $A \cap B = \{x | x \in A \wedge x \in B\}$.
- $A - B = \{x | x \in A \wedge x \notin B\}$.
- $\overline{A} = \{x | x \notin A\}$.

Example 4.26. *Use set builder notation and logical equivalences to prove that* $\overline{A \cup B} = \overline{A} \cap \overline{B}$.

Solution: We can prove this identity with the following steps.

$$\begin{aligned}
\overline{A \cup B} &= \{x | x \notin A \cup B\} \\
&= \{x | \neg(x \in A \cup B)\} \\
&= \{x | \neg(x \in A \vee x \in B)\} \\
&= \{x | \neg(x \in A) \wedge \neg(x \in B)\} \\
&= \{x | x \notin A \wedge x \notin B\} \\
&= \{x | x \in \overline{A} \wedge x \in \overline{B}\} \\
&= \{x | x \in \overline{A} \cap \overline{B}\} \\
&= \overline{A} \cap \overline{B}
\end{aligned}$$

Exercise: Use set builder notation and logical equivalences to prove that $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

Example 4.27. *Use set builder notation and logical equivalences to prove that* $A - B = A \cap \overline{B}$.

Solution: We can prove this identity with the following steps.

$$\begin{aligned}
A - B &= \{x | x \in A \wedge x \notin B\} \\
&= \{x | x \in A \wedge x \in \overline{B}\} \\
&= \{x | x \in A \cap \overline{B}\} \\
&= A \cap \overline{B}
\end{aligned}$$

Exercise: Use set builder notation and logical equivalences to prove that $\overline{A - B} = \overline{A} \cup B$.

Some Important Set Identities

1. Commutative laws: $A \cup B = B \cup A$ and $A \cap B = B \cap A$.
2. Associative laws: $A \cup (B \cup C) = (A \cup B) \cup C$ and $A \cap (B \cap C) = (A \cap B) \cap C$.
3. Distributive laws: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
4. De Morgan's laws: $\overline{A \cup B} = \overline{A} \cap \overline{B}$ and $\overline{A \cap B} = \overline{A} \cup \overline{B}$.
5. Absorption laws: $A \cup (A \cap B) = A$ and $A \cap (A \cup B) = A$.
6. Complement laws: $A \cup \overline{A} = U$ and $A \cap \overline{A} = \varnothing$.
7. Identity laws: $A \cap U = A$ and $A \cup \varnothing = A$.
8. Domination laws: $A \cup U = U$ and $A \cap \varnothing = \varnothing$.
9. Idempotent laws: $A \cup A = A$ and $A \cap A = A$.
10. Complementation law: $\overline{(\overline{A})} = A$.